



The Spring Test Profiler: Cut Down Your Test Execution Time by 70%

The hidden gem of Spring Test to unlock blazing fast feedback

J-Fall 2026 · November 12, 2026

Philip Riecks · @rieckpil · PragmaTech GmbH

Let's See Who Has the Fastest Build

📱 Scan the QR code or go to **menti.com** and enter

8999 9930



About Philip

- Software Engineer from Herzogenaurach (HQ of adidas & Puma), Germany 🍻
- Blogging and content creation about testing Java and Spring Boot applications 🍃
- Founder of PragmaTech GmbH - **Enabling Developers to Frequently Deliver** Software with **More Confidence**



Herzogenaurach, Bavaria

Why Test Software?

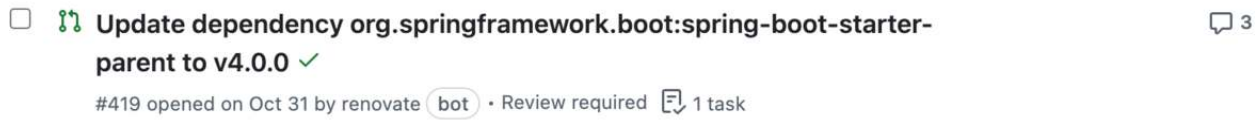


Most of the code is generated …

… but you still own the liability and can’t prompt confidence in your pipeline.

How Much Testing Do We Need?

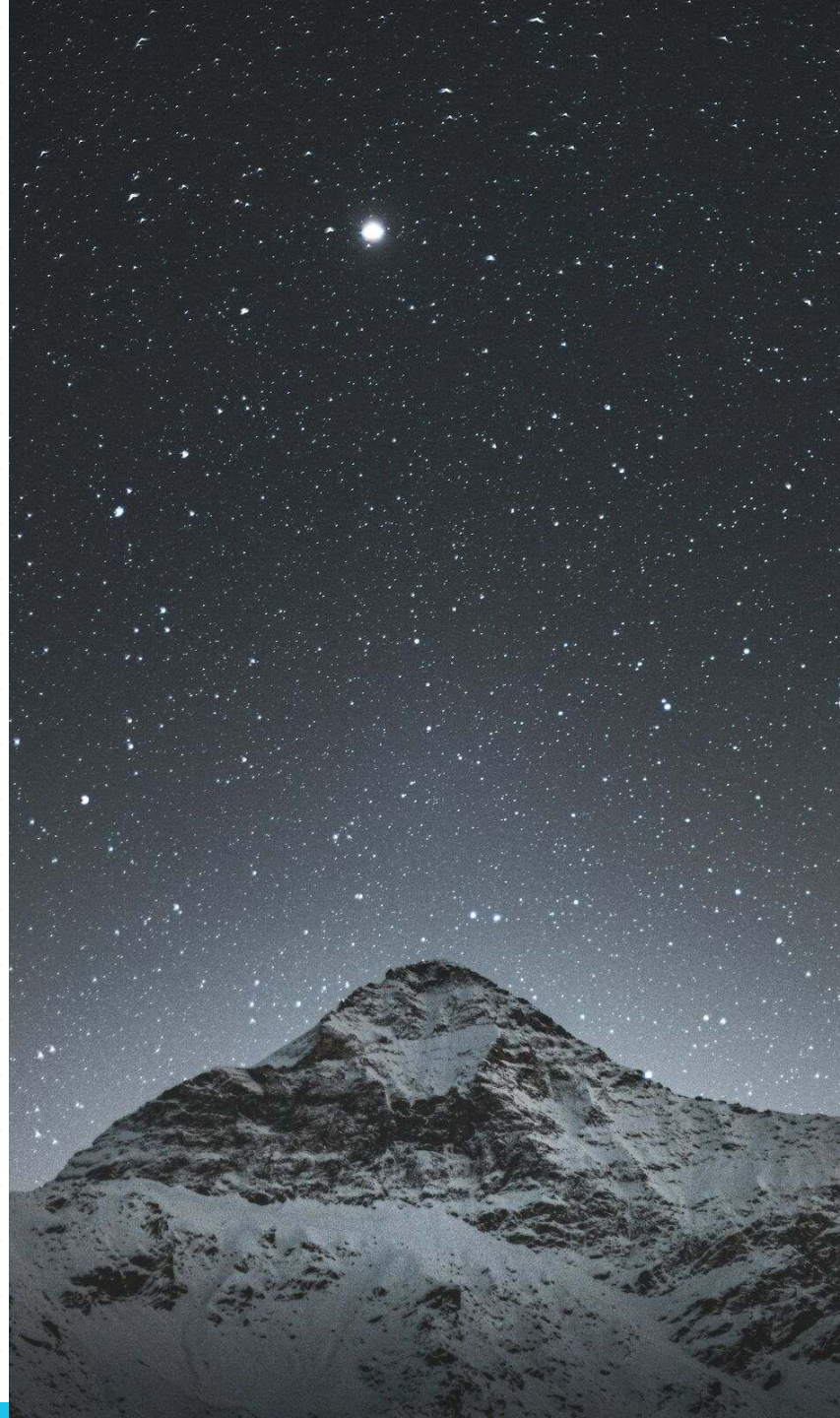
Imagine the following pull request:



How confident are you to deploy this to production on a Friday afternoon?

This requires **engineering excellence**:

- **Confidence** in every commit
- Being able to **deploy fast** anytime
- **Fast** feedback & **fast** flow



What We Expect From the Safety Net

Fast

Feedback in **minutes**, not hours

Comprehensive

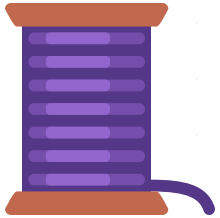
The **correct types of tests** to get confidence

Today we focus on fast tests.



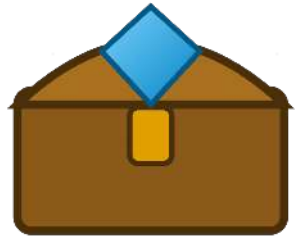
No tests at all

fastest build, zero confidence



Parallelize tests

use all your cores

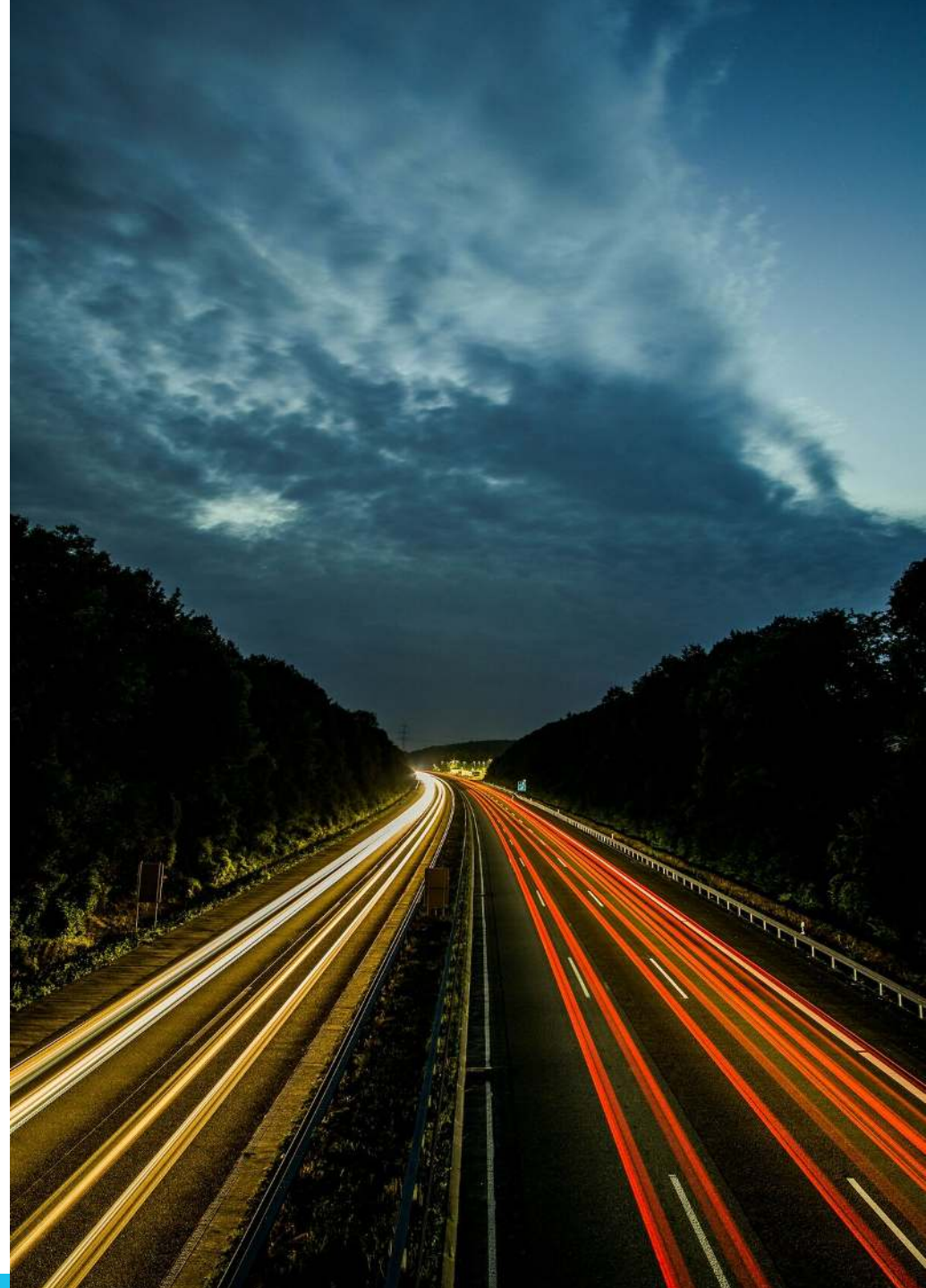


Something else ...

a treasure that helps us

THE SPRING TEST HIDDEN GEM

Introducing: Spring TestContext Context Caching



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What Do These Tests Share?

```java
@WebMvcTest(BookController.class)
class BookControllerIT {

  @Autowired
  private MockMvc mockMvc;

  @MockitoBean
  private BookService bookService;

  @Test
  void shouldReturnAllBooks() {
    // ...
  }
}
```

```java
@SpringBootTest(
  webEnvironment = RANDOM_PORT)
class BookApiIT {

  @Autowired
  private TestRestTemplate template;

  @Autowired
  private BookRepository repository;

  @Test
  void shouldCreateAndFetchBook() {
    // ...
  }
}
```

The Culprit: Every Test Starts a Context

```
  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/

 :: Spring Boot ::                (v4.1.1)

14:55:37.602 INFO  BookApiIT : Starting BookApiIT using Java 25.0.2
14:55:37.931 INFO  RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories
14:55:38.218 INFO  TomcatWebServer : Tomcat initialized with port 0 (http)
14:55:38.385 INFO  org.hibernate.orm.core : HHH000001: Hibernate ORM core version 7.4.5.Final
...
14:55:39.060 INFO  LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory
...
14:55:39.523 INFO  BookApiIT : Started BookApiIT in 2.034 seconds
```

To Avoid: Linear Build Time Growth



Understanding the Caching

1

Spring caches an already started context for later reuse

2

A test with a similar context configuration gets a "hot" context handed over

3

No dependency needed, it works out of the box

How Does Spring Cache Your Context?

Step 1: Implement a test with a context

```
@SpringBootTest
class BookApiIT {
  // ...
}
```

Step 2: Spring analyzes your context requirements

Profiles: default
Beans: mocks, spies
Properties:
server.port=0
...

Step 3: Computation of the cache key and trying to look up an existing context

Cache hit: Reuse hot context
Cache miss: Your test needs to start the context

Each Test Requires a Different Context

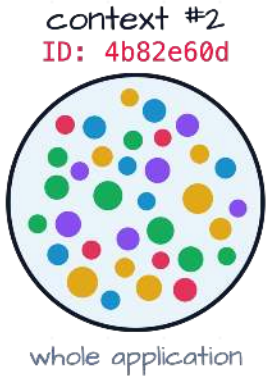
```java
@WebMvcTest(BookController.class)
class BookControllerIT {
  @MockitoBean BookService bookService;
  // ...
}
```






```java
@SpringBootTest(webEnvironment = RANDOM_PORT)
class BookApiIT {
  @Autowired TestRestTemplate template;
  // ...
}
```





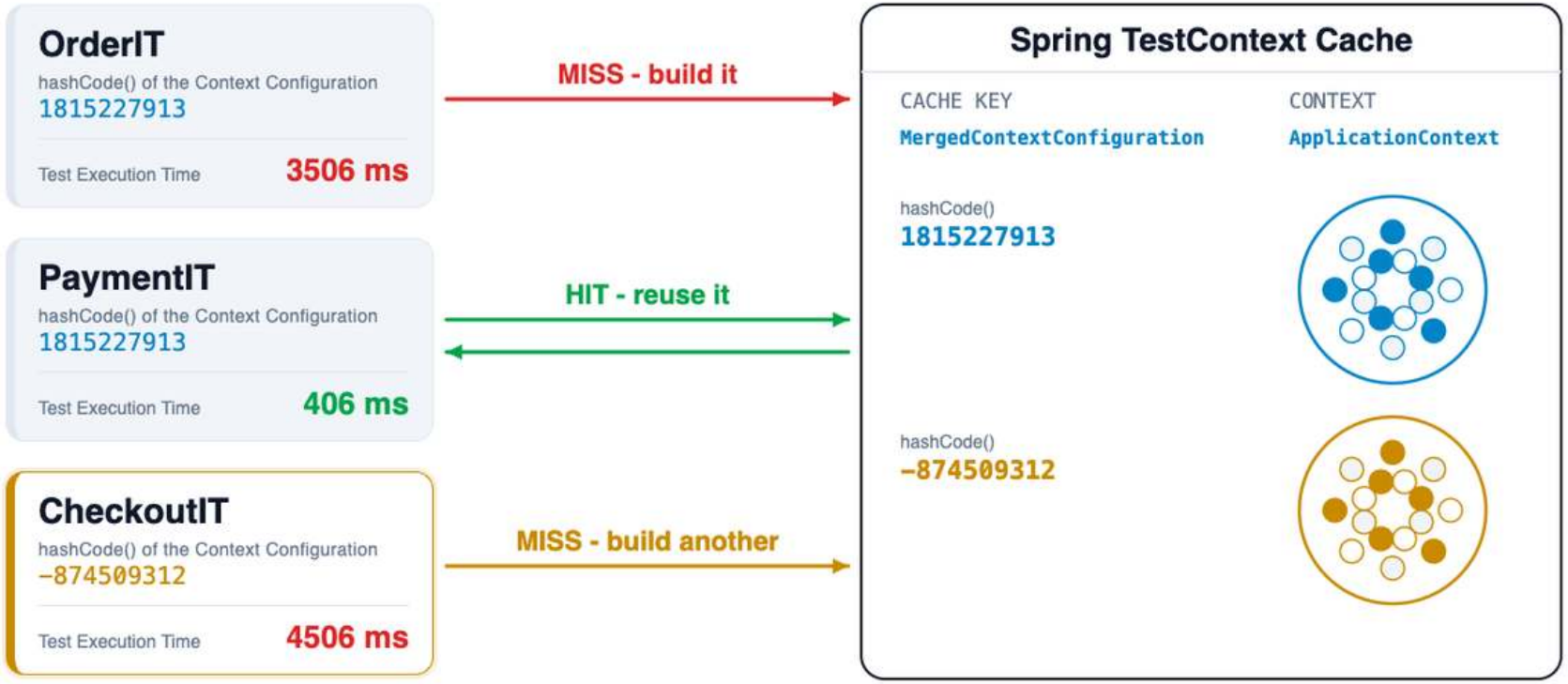
Spring TestContext Caching Behind the Scenes

```
1   // DefaultContextCache.java
2   private final Map<MergedContextConfiguration, ApplicationContext> contextMap =
3     Collections.synchronizedMap(new LruCache(32, 0.75f));
```

The cache key is a `MergedContextConfiguration`: **the store of all context customization points** of a test class.

- Spring collects **more than 10 things** to identify a context
- Any change results in a **new hash code** of this Java object
- A new hash code means a cache miss and a new context start

Context Caching in Action



Result: 3 tests ran, but 2 contexts had to start

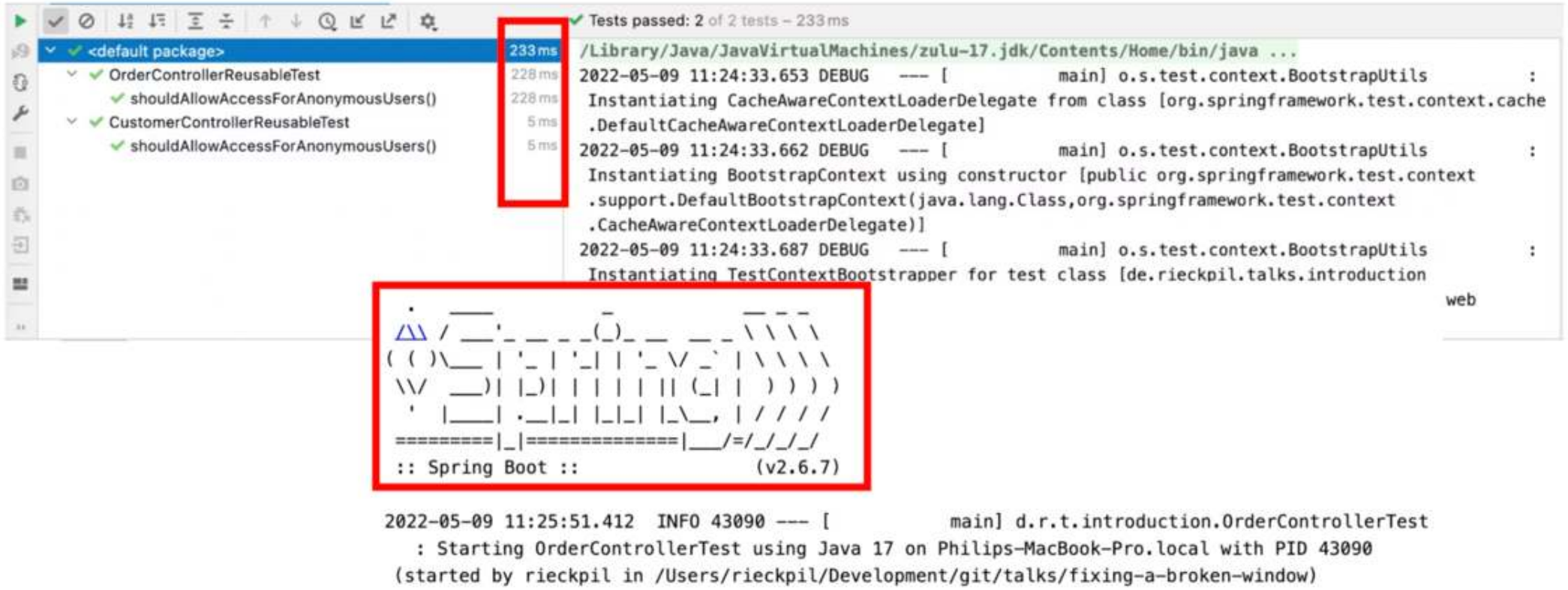
Only **PaymentIT** reused a cached context and finished fast - the other two each paid for a fresh one.

Context Killers

Anything that changes the **cache key** forces a **new context start**:

- `@MockitoBean` / `@MockitoSpyBean` on single test classes
- `@TestPropertySource`, `properties = ...` or `@DynamicPropertySource` per class
- Different `@ActiveProfiles`
- Different `@Import`, `@TestConfiguration` or `@ContextConfiguration`
- Mixing `@SpringBootTest` with slices like `@WebMvcTest` or `@DataJpaTest`
- Different `webEnvironment` (`MOCK` vs. `RANDOM_PORT`)
- `@DirtiesContext` - the final boss: **removes** the context from the cache

Detect Context Restarts - Visually



Detect Context Restarts - with Logs

```
<logger name="org.springframework.test.context" level="TRACE" />
```

```
2022-05-05 14:27:38.136 DEBUG 42500 --- [           main] org.springframework.test.context.cache   : Spring test
 ApplicationContext cache statistics: [DefaultContextCache@68e62b3b size = 1, maxSize = 32, parentContextCount = 0,
 hitCount = 11, missCount = 1]
2022-05-05 14:27:38.136 DEBUG 42500 --- [           main] tractDirtiesContextTestExecutionListener : After test class:
 context [DefaultTestContext@325bb9a6 testClass = ApplicationIT, testInstance = [null], testMethod = [null], testException
  = [null], mergedContextConfiguration = [WebMergedContextConfiguration@1d12b024 testClass = ApplicationIT, locations =
 '{}', classes = '{class de.rieckpil.talks.Application}', contextInitializerClasses = '[]', activeProfiles = '{}',
 propertySourceLocations = '{}', propertySourceProperties = '{org.springframework.boot.test.context
 .SpringBootTestContextBootstrapper=true, server.port=0}', contextCustomizers = set[org.springframework.boot.test.context
 .filter.ExcludeFilterContextCustomizer@5215cd9a, org.springframework.boot.test.json
 .DuplicateJsonObjectContextCustomizerFactory$DuplicateJsonObjectContextCustomizer@9257031, org.springframework.boot.test
```

Detect Context Restarts - with Tooling



An open-source Spring Test utility that provides visualization and insights for Spring Test execution, with a focus on Spring context caching statistics.

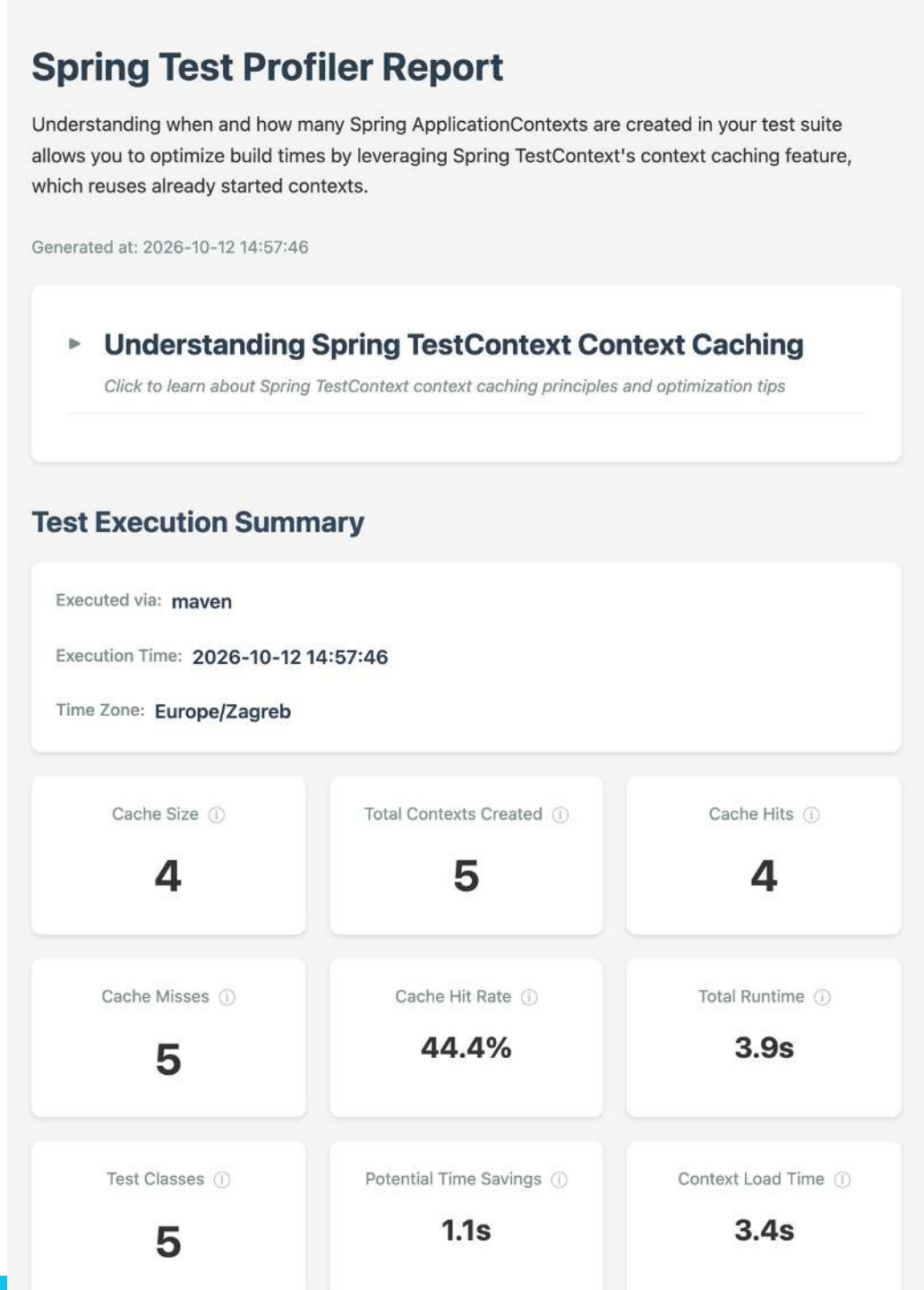
Overall goal: Identify optimization opportunities in your Spring Test suite to speed up your builds and ship to production faster and with more confidence.

SPRING TEST PROFILER

Demo Time

A quick look at the report.

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Spring Test Profiler Report

Understanding when and how many Spring ApplicationContexts are created in your test suite allows you to optimize build times by leveraging Spring TestContext's context caching feature, which reuses already started contexts.

Generated at: 2026-10-12 14:57:46

▸ **Understanding Spring TestContext Context Caching**

Click to learn about Spring TestContext context caching principles and optimization tips

Test Execution Summary

Executed via: **maven**

Execution Time: **2026-10-12 14:57:46**

Time Zone: **Europe/Zagreb**

Cache Size	Total Contexts Created	Cache Hits
4	5	4

Cache Misses	Cache Hit Rate	Total Runtime
5	44.4%	3.9s

Test Classes	Potential Time Savings	Context Load Time
5	1.1s	3.4s

How to Cut Build Time by 70%

1

Understand

Measure the context starts with the Spring Test Profiler

2

Optimize

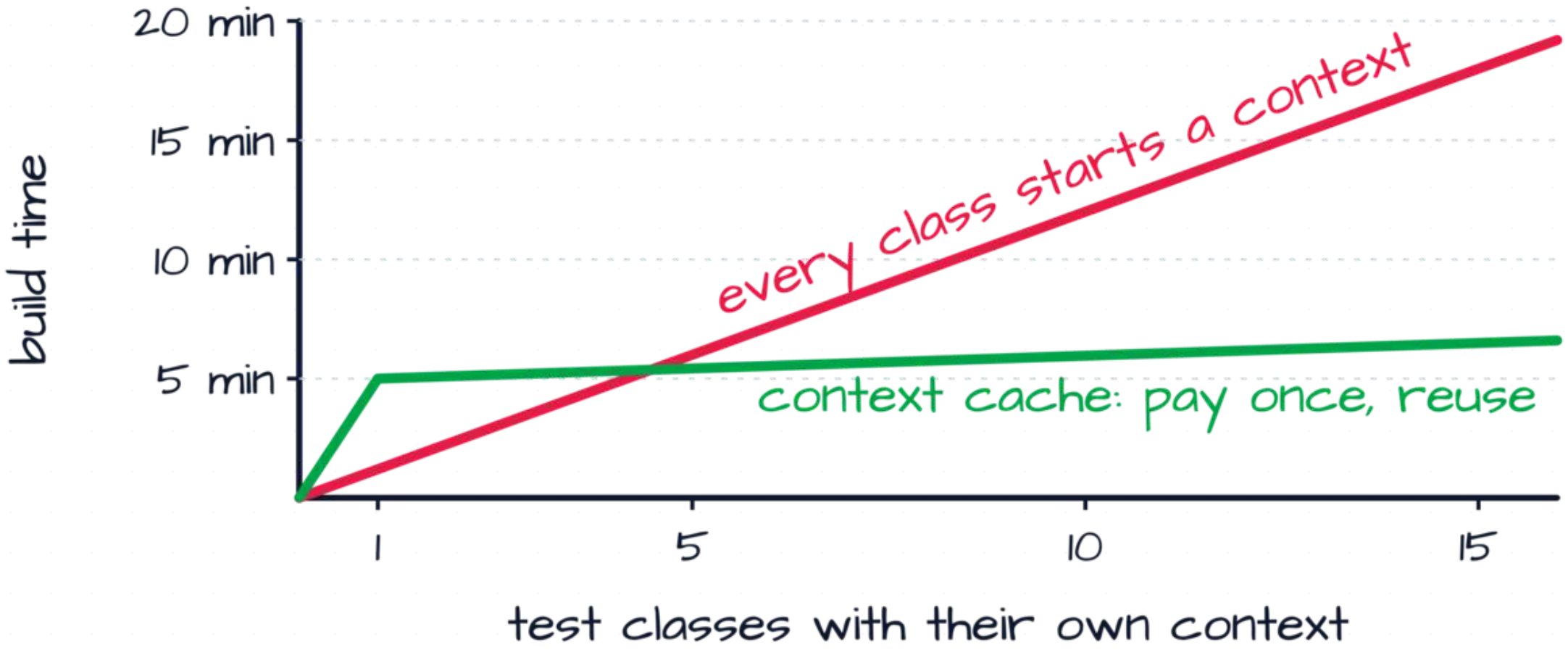
Align the configurations so contexts get reused

3

Ensure

Keep it fast, for your team and for your AI agents

The Goal: Steady Build Times



Real Results: Builds I Optimized at Clients

26 → 12 min

build time

about 54% faster

15 → 3 min

build time

about 80% faster

Add the Spring Test Profiler

An open-source Spring Test utility that visualizes context caching for your test suite:

```
<dependency>
  <groupId>digital.pragmatech.testing</groupId>
  <artifactId>spring-test-profiler</artifactId>
  <version>0.5.0</version>
  <scope>test</scope>
</dependency>
```

Nothing to Activate

No setup needed: just drop the dependency and run your build.

Then open the report:

```
./mvnw verify
open target/spring-test-profiler/latest.html
```

Don’t Customize the Context Heavily

- Every `@MockitoBean`, `@MockitoSpyBean`, `@TestPropertySource` or `@ActiveProfiles` variation is **a new context**
- Keep **one shared base class** and one shared test configuration
- Put mocks into shared configuration instead of annotating single test classes
- Slice only where it really pays off, and align the slices as well

The Fix: One Base Class

```
1  @SpringBootTest(webEnvironment = RANDOM_PORT)
2  @AutoConfigureTestRestTemplate
3  abstract class AbstractIntegrationTest {
4  }
5
6  class BookApiIT
7      extends AbstractIntegrationTest {
8    // no own annotations anymore
9  }
```

- One annotation set for all tests
- No `@MockitoBean` or `@MockitoSpyBean`
- No extra `properties`

The Final Boss

Developers tend to consult AI/StackOverflow for integration test issues and often copy advice from the internet without knowing the implications:

```
1   @SpringBootTest
2   @DirtiesContext
3   // this instructs Spring to remove the context from the cache
4   // and rebuild a new context on every request
5   public abstract class AbstractIntegrationTest {
6
7   }
```

The setup above will **disable** the context caching feature and slow down the builds significantly!

Educate Your Team

- Explain the **cache key**: every new annotation combination costs a context start
- Put shared configuration into **one base class**, and review changes to it
- Treat `@DirtiesContext` and a new `@MockitoBean` like a performance decision
- Don't religiously optimize for only **one context**, a handful works

Educate Your AI Agents Too

Agents write most of the tests now. They need the same rules:

```
.claude/skills/
├── unit-testing              fast tests without context bloat
├── slice-testing             right-sized Spring context slices
├── integration-testing       extend AbstractIntegrationTest, no new context
└── test-setup-reviewer       flags context cache killers
```

Point the skill to your `AbstractIntegrationTest` and name the antipatterns.

The Profiler Also Writes a JSON Report

`target/spring-test-profiler/results.json` - flat and easy to consume from a pipeline:

```
{
  "schemaVersion": 1,
  "profilerVersion": "0.5.0",
  "generatedAt": "2026-10-12T12:57:46Z",
  "buildTool": "MAVEN",
  ...
  "contextsCreated": 5,
  "contextCacheHits": 4,
  "contextCacheMisses": 5,
  "contextCacheHitRatio": 0.4444,
  ...
}
```

Let the CI/CD Pipeline Guard the Context Count

Add one step at the end of the pipeline that reads the report:

```
./mvnw verify

expectedContexts=5
actualContexts=$(jq -r '.contextsCreated' \
  target/spring-test-profiler/results.json)

if [ "$actualContexts" -gt "$expectedContexts" ]; then
  echo "Too many Spring contexts: $actualContexts"
  exit 1
fi
```

Upcoming: Guard the Context Count with a Test

The profiler exposes the numbers to your own code (upcoming feature):

```
1   public class ContextBudgetListener implements LauncherSessionListener {
2
3     @Override
4     public void launcherSessionClosed(LauncherSession session) {
5       var statistics = SpringTestProfilerListener.getContextCacheStatistics();
6
7       if (statistics.distinctContextsCreated() > 5) {
8         throw new AssertionError("Too many Spring contexts: "
9             + statistics.describeContexts());
10      }
11    }
12  }
```

Summary



The problem

Every context start costs time, so build time grows with every test class



The gem

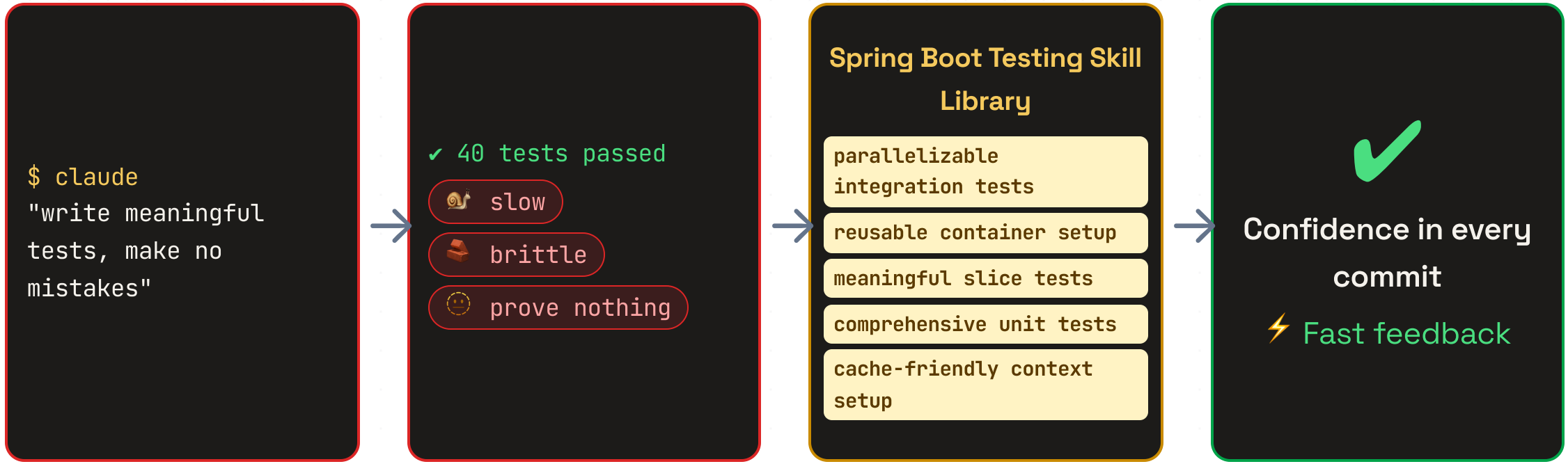
Context caching: same configuration, same context



The plan

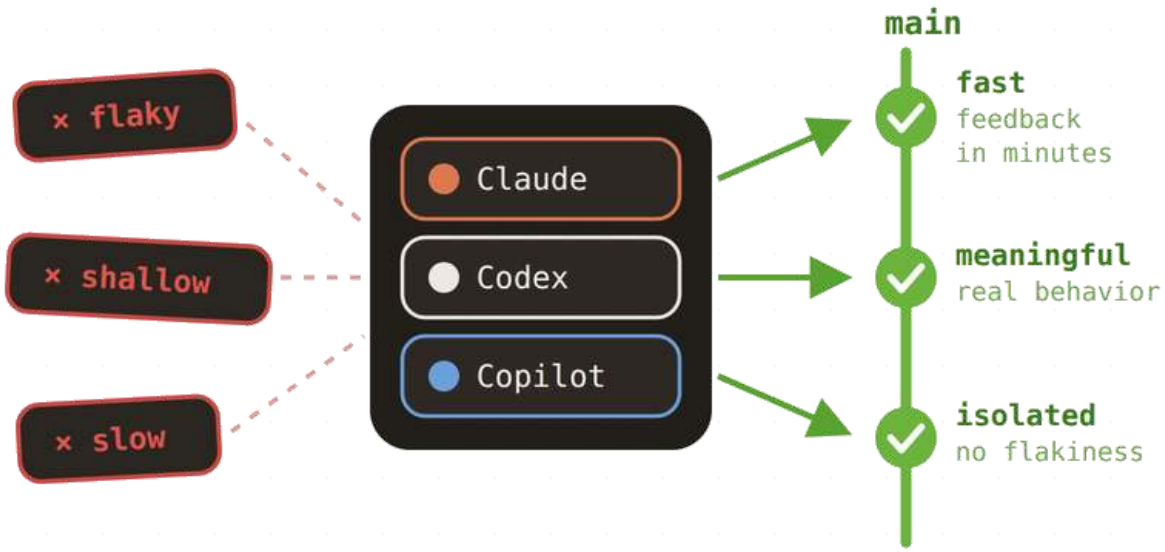
Understand with the profiler, optimize, ensure with CI and skills

From Years of Testing Knowledge to a Skill Library



Agentic Testing for Spring Boot

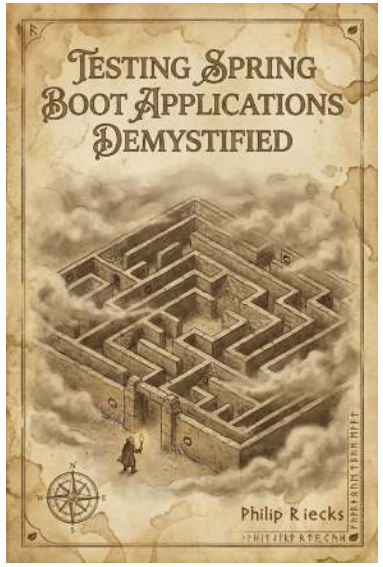
Onboard your team to the skill library: make Claude, Codex and Copilot write tests **a lead Spring Boot dev would have written**.



DM me on LinkedIn (Philip Riecks) to get the details.

Bedankt and Joyful Testing!

Join my **Spring Boot Testing Newsletter** for tips and get **Testing Spring Boot Applications Demystified** for free:



rieckpil.de/free-spring-boot-testing-book

